

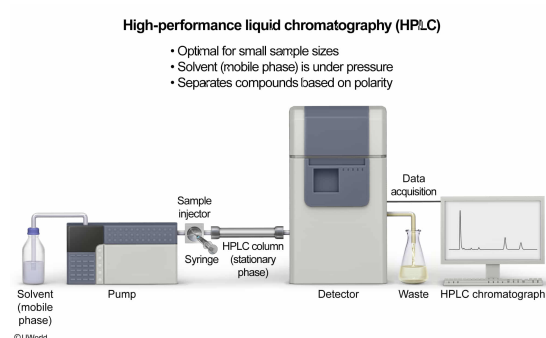
Which separation technique is optimal for purification of small sample sizes and employs a stationary phase, a liquid mobile phase (under pressure), and a detector to separate compound mixtures based on polarity?

- ☐ A. Gas chromatography [8%]
- ☐ B. Extraction [1%]
- ✓ ☒ C. High-performance liquid chromatography [73%]
- ☐ D. Thin-layer chromatography [16%]

Correct

72% answered correctly

Explanation:



Organic reactions generally do not produce a single product; therefore, separation and purification methods are imperative to obtain a compound of interest. Several techniques exist to separate compounds based on their physical properties, including boiling point, solubility, and polarity.

High-performance liquid chromatography (HPLC) is a purification technique comprised of a mobile phase, stationary phase, detector, and computer for data acquisition. The **mobile phase** is a **liquid solvent** or mixture of solvents pumped through the system under **pressure**. The stationary phase is a column made of either a polar (**normal-phase**) or nonpolar (**reversed-phase**) material, depending on the nature of the compounds to be separated.

HPLC is optimal for small sample sizes as it **separates compounds** based on their **polarity**, giving each compound a different affinity for the mobile and stationary phases and causing the compounds to pass through the column at different rates. Compounds are detected as they come off the column and go into the waste or are collected separately. Data is transmitted to the computer, creating a chromatograph.

(Choice A) **Gas chromatography** requires small sample sizes but separates compounds based on boiling point. This method contains a stationary and mobile phase; however, the mobile phase during separation is a *gas* rather than a liquid solvent.

(Choice B) **Extraction** separates compounds based on their relative solubility in different solvents. This method uses two immiscible solvents that make up the aqueous and organic layers; no stationary phase, mobile phase, or detector is required for extraction.

(Choice D) **Thin-layer chromatography (TLC)** separates small sample sizes and includes a

stationary phase, solvent mobile phase, and detector (UV). However, TLC is not performed under pressure.

Educational objective:

High-performance liquid chromatography is a purification technique that separates compounds based on polarity and consists of a mobile phase, stationary phase, detector, and computer for data acquisition. The mobile phase is a solvent or mixture of solvents that are pumped through the system under pressure, and the stationary phase is a column made of either a polar or nonpolar material.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods